Homework 5 : Optional Bonus

This deliverable is an optional extra credit assignment, it will be graded as usual, with the standard late penalties,

Problem-1

1. For each example, state whether or not the censoring mechanism is independent. Justify your answer.

   (a) In a study of disease relapse, due to a careless research scientist, all patients whose phone numbers begin with the number "2" are lost to follow up.

   (b) In a study of longevity, a formatting error causes all patient ages that exceed 99 years to be lost (i.e. we know that those patients are more than 99 years old, but we do not know their exact ages).

   (c) Hospital A conducts a study of longevity. However, very sick patients tend to be transferred to Hospital B, and are lost to follow up.

   (d) In a study of unemployment duration, the people who find work earlier are less motivated to stay in touch with study investigators, and therefore are more likely to be lost to follow up.

   (e) In a study of pregnancy duration, women who deliver their babies pre-term are more likely to do so away from their usual hospital, and thus are more likely to be censored, relative to women who deliver full-term babies.

   (f) A researcher wishes to model the number of years of education of the residents of a small town. Residents who enroll in college out of town are more likely to be lost to follow up, and are also more likely to attend graduate school, relative to those who attend college in town.

   (g) Researchers conduct a study of disease-free survival (i.e. time until disease relapse following treatment). Patients who have not relapsed within five years are considered to be cured, and thus their survival time is censored at five years.

Problem-2

We conduct a study with $n = 4$ participants who have just purchased cell phones, in order to model the time until phone replacement. The first participant replaces her phone after 1.2 years. The second participant still has not replaced her phone at the end of the two-year study period. The third participant changes her phone number and is lost to follow up (but has not yet replaced her phone) 1.5 years into the study. The fourth participant replaces her phone after 0.2 years.

For each of the four participants ($i = 1, \ldots, 4$), answer the following questions using the notation introduced in Section 11.1:

(a) Is the participant's cell phone replacement time censored?

(b) Is the value of c_i known, and if so, then what is it?

(c) Is the value of t_i known, and if so, then what is it?

(d) Is the value of y_i known, and if so, then what is it?

(e) Is the value of δ_i known, and if so, then what is it?

Problem-3

This exercise focuses on the brain tumor data, which is included in the `ISLR2` `R` library.

(a) Plot the Kaplan-Meier survival curve with ± 1 standard error bands, using the `survfit()` function in the `survival` package.

(b) Draw a bootstrap sample of size $n = 88$ from the pairs (y_i, δ_i), and compute the resulting Kaplan-Meier survival curve. Repeat this process $B = 200$ times. Use the results to obtain an estimate of the standard error of the Kaplan-Meier survival curve at each timepoint. Compare this to the standard errors obtained in (a).

(c) Fit a Cox proportional hazards model that uses all of the predictors to predict survival. Summarize the main findings.

(d) Stratify the data by the value of `ki`. (Since only one observation has `ki=40`, you can group that observation together with the observations that have `ki=60`.) Plot Kaplan-Meier survival curves for each of the five strata, adjusted for the other predictors.

Problem-4

Observation (Y)	Censoring Indicator (δ)	Covariate (X)
26.5	1	0.1
37.2	1	11
57.3	1	-0.3
90.8	0	2.8
20.2	0	1.8
89.8	0	0.4

TABLE 11.4. *Data used in Exercise 4.*

This example makes use of the data in Table 11.4.

(a) Create two groups of observations. In Group 1, $X < 2$, whereas in Group 2, $X \geq 2$. Plot the Kaplan-Meier survival curves corresponding to the two groups. Be sure to label the curves so that it is clear which curve corresponds to which group. By eye, does there appear to be a difference between the two groups' survival curves?

(b) Fit Cox's proportional hazards model, using the group indicator as a covariate. What is the estimated coefficient? Write a sentence providing the interpretation of this coefficient, in terms of the hazard or the instantaneous probability of the event. Is there evidence that the true coefficient value is non-zero?

(c) Recall from Section 11.5.2 that in the case of a single binary covariate, the log-rank test statistic should be identical to the score statistic for the Cox model. Conduct a log-rank test to determine whether there is a difference between the survival curves for the two groups. How does the p-value for the log-rank test statistic compare to the p-value for the score statistic for the Cox model from (b)?